



A fast calibration algorithm for Non-Dispersive Infrared single channel carbon dioxide sensor based on deep learning[☆]

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ABSTRACT

As people pay more attention to environmental monitoring, Carbon dioxide (CO₂) sensors are widely used. However, most of the infrared CO₂ single-channel sensors are accompanied by low calibration efficiency and low accuracy. In order to save costs while improving calibration efficiency and accuracy, we proposed a fast calibration algorithm for Non-Dispersive Infrared (NDIR) single-channel carbon dioxide sensor based on deep learning. Firstly, we establish N network models which consist of N sensors by collecting m data points from different temperatures and concentrations. Secondly, we collect six data points from a new sensor which are measured at three temperatures and two concentrations. Thirdly, we choose multiple approximate models from N network models based on the matching of the data points. At last, we regard these models as the estimation model of the new sensor to calibrate the sensor concentration. This method eliminates the individual differences of a single model to a certain extent and achieves the purpose of rapid calibration. After comparing three kinds of neural networks and conducting relevant experiments, we chose BP neural network as the model, and set the number of selected models to three. The results show that the floating up and down by industry-standard 5% plus or minus 50 ppm calculation, the qualified rate of our method is up to 91.542% between 0 °C to 45 °C, and the qualified rate even reaches 99.063% between 20 °C to 35 °C. Compared with similar products, the qualified rate of our method in the calibration of carbon dioxide increases by 12.315% and 22.732% respectively.

1. Introduction

CO₂ is one of the main components of the atmosphere and one of the main gases that cause the greenhouse effect [1]. A large amount of CO₂ emissions will cause serious environmental problems. Using carbon dioxide sensors is one of the most direct methods to measure carbon dioxide gas concentration [2]. And they are widely used in environmental monitoring. With the maturity of sensor technology, many places adopt the form of a sensor network to monitor the city's greenhouse gas emissions [3–6]. The evaluation and application of several low-cost sensors have shown their promise in environmental monitoring [7–9]. At present, the types of carbon dioxide sensors include resistance type, catalytic combustion type, electrochemical type, optical type, thermal conductivity type, etc. Carbon dioxide sensors based on NDIR technology have been widely used because of their wide measurement range, high sensitivity, high accuracy, fast response, poisoning resistance and long life [10,11].

Carbon dioxide sensors based on NDIR technology need to be calibrated before using. Calibration is to measure and test the sensor, and to establish the corresponding relationship or model between the sensor test voltage value and CO₂ concentration. However, in addition to the influence of CO₂ concentration itself on the voltage value, factors such as ambient temperature, pressure, sensor manufacturing technology, and light source stability also have a certain impact on the output voltage value of the sensor. The measured concentration output of the sensor may be discrepant with the actual concentration. Therefore, accurate calibration of carbon dioxide sensor can provide the basic guarantee for accurate measurement of CO₂ concentration. Carbon dioxide sensors for accurate calibration need to spend a lot of manpower, financial resources and time cost. Therefore, to achieve a highly efficient, accurate and low-cost calibration method has a good application value and market prospects.

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Non-dispersive infrared carbon dioxide sensors can be divided into single-channel and dual-channel. The cost of single-channel carbon dioxide sensors cost is less than dual-channel carbon dioxide sensors. In this paper, based on NDIR technology and the idea that there are commonalities among different gas sensors, we proposed an efficient, low-cost and highly reliable fast calibration algorithm for single-channel carbon dioxide sensors. In the same material and formation process, the data model established by each gas sensor also has some similarities. We have established a certain number of relevant data models, it can estimate the data model of the newly produced sensor through the relevant data models and the measurement data of the newly produced sensor, so as to realize the sensor calibration of the newly produced sensor.

The main contributions of this paper are as follows:

(1) This paper proposes a fast calibration algorithm based on deep learning. The algorithm uses the existing model and a small number of data points to build a new sensor model, thus saving a lot of time of measuring data. It not only saves various costs, but also has higher efficiency and accuracy.

(2) According to industry standards, our algorithm can achieve a data point qualified rate of 91.542% at 0 °C–45 °C and a data point qualified rate of 99.063% at 20 °C–35 °C. Compared with the two competing products, the average qualified rate of the algorithm data points have increased by 12.315% and 22.732% respectively.

The rest of this paper is structured as follows. Section 2 reviews the related work of carbon dioxide sensors based on NDIR. Section 3 describes the relevant principles involved in the algorithm. Section 4 describes the three type of model used in the algorithm. Section 5 presents the specific design of the algorithm. In Section 6, the performance of the proposed method is experimentally evaluated, and Section 7 concludes this paper.

2. Related work

As technology matures and commercialization develops, most products are moving towards low-cost development. In the literature [12, 13], in order to reduce the cost of CO₂ monitoring in the environment, a dual-channel sensor is adopted and the sensor is optimized. Although the optimization results were significant, the cost of the sensors themselves did not decrease. Part of the reason for good optimization is that dual-channel sensors have been able to eliminate some of the environmental impacts to a certain extent. Compared with the dual-channel sensor, the single-channel sensor has a lower cost.

In terms of calibration of carbon dioxide sensor, multiple linear regression method was used in the literature [12,14,15], random forest model was used in the literature [16], and artificial neural network model was used in the literature [15]. The existing literature mainly aims to obtain sensor models with better performance, and most of the literature only optimizes the dual-channel sensor to replace the gas monitor. The cost of the sensors by these methods is much cheaper than gas monitors, and it can increase the range of monitoring by these methods, but it is much more expensive than single-channel carbon dioxide sensors.

In the literature [17,18], the neural network model was used to calibrate the carbon dioxide sensor. But when building a sensor model for the new sensor, it needs to collect 40 points, and each data point needs to take 30 to 40 min. It takes about 24 h for modeling the sensor, and it require a large amount of gas for 40 data points. However, the algorithm designed by us only needs to collect six data points for the newly produced sensor. Hence, our algorithm dramatically reduces the financial cost and time cost.

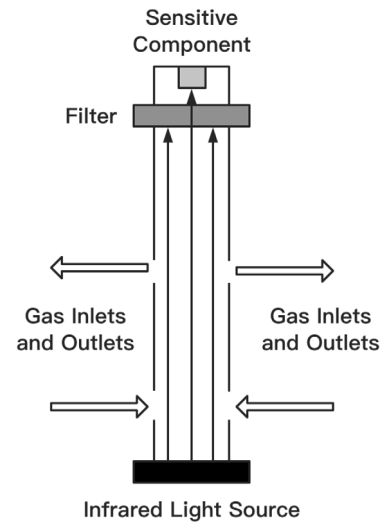


Fig. 1. Simple schematic diagram of sensor structure.

3. Principle of infrared gas sensor

The simple structure diagram of a single-channel infrared gas sensor is shown in Fig. 1, which mainly includes an infrared light source, filter, sensitive component, gas inlets and outlets.

Most gases have a specific absorption wavelength in the non-dispersive infrared wavelength range, and the absorption peak of CO₂ is located at 4.26 μm [19,20]. The absorption of infrared radiation by CO₂ satisfies the Lambert–Beer law [21]:

$$I = I_0 \exp(-kCL) \quad (1)$$

Among them, I_0 is the intensity of incident light, I is the intensity of emitted light, k is the absorption coefficient of CO₂, L is the effective optical length of CO₂ absorption, and C is the concentration of CO₂.

Theoretically, the effective wavelength of light absorbed by CO₂ in the sensor is a fixed value. In the same environmental conditions, the absorption coefficient k is basically a fixed value, and we just need to know the incident light intensity I_0 and emission intensity I , then it can calculate the concentration C of CO₂. However, it is influenced by some factors such as temperature and pressure in a real environment, absorption coefficient k is very complex and changeable, so it is difficult to maintain a constant state. If it directly use Eq. (1), it is very difficult to obtain the concentration. To a certain extent, it can reduce the influence of environmental factors with a dual-channel carbon dioxide sensor, but the manufacturing cost of a dual-channel sensor is much higher than a single-channel sensor. Therefore, we need to take measures to eliminate the impact of these environmental factors at a low cost.

4. System models

This paper mainly uses three types of model for experimental comparison as follows: BP neural network, LSTM neural network and GRU neural network. In this algorithm, the purpose of using the neural network is to establish a model of the sensor, which is similar to a curved surface and has a good sense of continuity. BP neural network, LSTM neural network and GRU neural network are all suitable for this problem, and each has its advantages.

In order to make the model converge during training, the input value and label value (i.e. the real CO₂ concentration value) during training are normalized, and the normalization formula is:

$$X^* = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (2)$$

X^* is the normalized value. X is the actual value. X_{min} is the theoretical minimum. And X_{max} is the theoretical maximum.

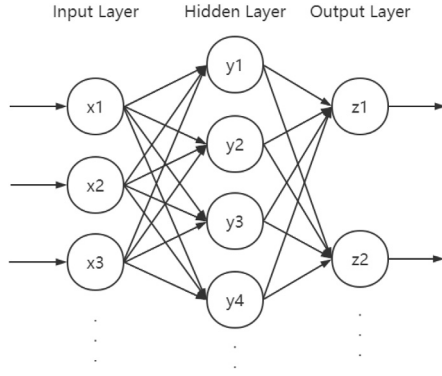


Fig. 2. Schematic diagram of BP neural network structure.

4.1. BP neural network

BP neural network is a feedforward neural network based on BP algorithm [22]. It includes input layer, hidden layer, and output layer, which is as shown in Fig. 2. BP neural network can learn and store a large number of mapping relations between input and output models, and it does not need to know the mathematical equations of these mapping relations in advance [23,24]. The original output voltage and temperature in the cavity are used as input values, and the gas distribution parameters of the experimental equipment are used as label values to train the BP neural network model. It can eliminate the influence of environmental factors to a certain extent, and make the measured value closer to the actual value. And in terms of the complexity of the sensor model, the BP neural network is sufficient, which will be explained in subsequent experiments.

The BP neural network used in this paper is a three-layer network model. There is an input layer, a hidden layer, and an output layer. The sensor output voltage (*AD value*) and the sensor cavity temperature (*T*) are used as input values. The hidden layer uses 7 neurons. The output value is the concentration value (*C*) of CO₂. The *hyperbolic tangent function* is used as the activation function between the input layer and the hidden layer. The *sigmoid* function is used as the activation function between the hidden layer and the output layer.

4.2. LSTM neural network

LSTM (Long short-term memory) is a special Recurrent Neural Network (RNN) [25]. The neural network structure of LSTM is the same as that of ordinary RNN, and its structure is shown in Fig. 3. RNN allows the information of the previous step to be passed to the next step. But as the number of cycles increases, RNN cannot effectively use historical information. To solve this problem, LSTM was born. Its main function is to solve the problem of gradient disappearance and gradient explosion during long sequence training. Compared with RNN, LSTM can perform better in longer sequences. In this problem, the measurement of carbon dioxide concentration is a long-term continuous process. The use of LSTM can save memory information during the process of using and reduce the impact of environmental factors. The specific advantages and disadvantages need to be concluded in the experimental process. The main difference between LSTM and RNN is the structure of block A. The main structure of block A is shown in Fig. 4. It mainly includes three gates (forget gate, input gate, output gate) [26,27].

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (3)$$

Part 1 is the cell state, which is equivalent to a conveyor belt. It transfers the control information at this moment to the next moment. This control information is C_t . The update method of C_t is shown in Eq. (3). C_{t-1} is the control information of the previous step, and other

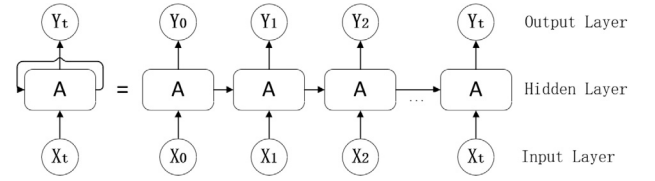


Fig. 3. Schematic diagram of RNN structure.

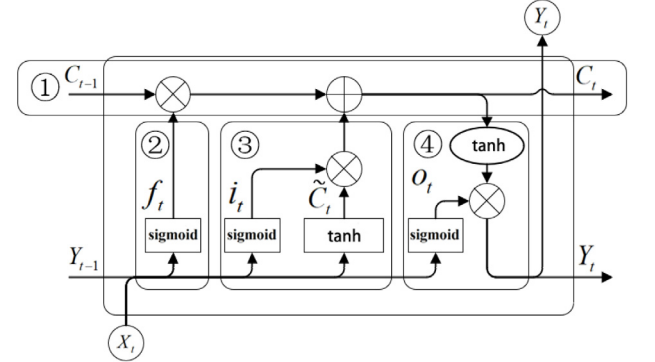


Fig. 4. Schematic diagram of LSTM structure.

parameters will be introduced below. C_t will be transmitted all the time to achieve the purpose of memory.

$$f_t = \sigma(W_{fy} \cdot Y_{t-1} + W_{fx} \cdot X_t + b_f) \quad (4)$$

Part 2 is the forget gate, which consists of a *sigmoid* layer. The *sigmoid* layer outputs a vector, and each number in the vector is between 0 and 1. The closer the number is to 1, the more the information is accepted, and the 1 represents the complete acceptance; the closer the number is to 0, the more the information is forgotten, and the 0 represents complete forgotten. It can determine what information can pass through the cell state. This vector is f_t , and its calculation formula is shown in Eq. (4). This vector is obtained by multiplying the output value (Y_{t-1}) of the previous step and the input value (X_t) of this step by the weight W_f of this part and then performing the *sigmoid* operation.

$$i_t = \sigma(W_{iy} \cdot Y_{t-1} + W_{ix} \cdot X_t + b_i) \quad (5)$$

$$\tilde{C}_t = \tanh(W_{cy} \cdot Y_{t-1} + W_{cx} \cdot X_t + b_c)$$

Part 3 is the input gate, which is composed of a *sigmoid* layer, a *tanh* layer and an element-wise multiplication operation. The *tanh* layer generates new candidate values, and the *sigmoid* layer determines which of the candidate values are used for updating. Finally, the generated value is added to the cell state. The specific calculation steps are shown in Eq. (5). i_t is the output value of the *sigmoid* layer, which is obtained by multiplying the output value of the previous step (Y_{t-1}) and the input value of this step (X_t) by the weight of this layer (W_i), and then performing the *sigmoid* operation; $\tilde{C}_t$ is the output value of the *tanh* layer, which is obtained by multiplying the output value (Y_{t-1}) of the previous step and the input value (X_t) of this step by the weight (W_c) of this layer, and then performing the *tanh* operation. Finally, an element-wise multiplication operation is performed on the two output values.

$$o_t = \sigma(W_{oy} \cdot Y_{t-1} + W_{ox} \cdot X_t + b_o) \quad (6)$$

$$Y_t = o_t \cdot \tanh(C_t)$$

Part 4 is the output gate, which consists of a *sigmoid* layer, an element-wise multiplication operation and a *tanh* operation. The purpose of the *tanh* operation is to compress the previously learned information to stabilize the value. The *sigmoid* layer obtains the output of information without considering the information previously learned.

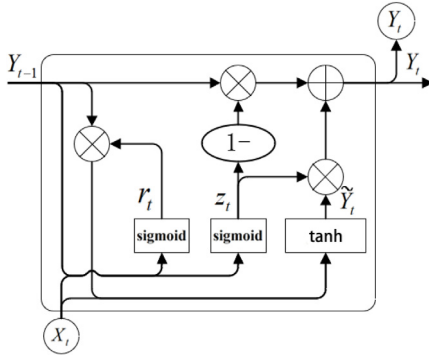


Fig. 5. Schematic diagram of GRU structure.

Finally, it will get the output value by multiplying the two in pairs. The specific calculation steps are shown in Eq. (6). o_t is the output value of the *sigmoid* layer, which is obtained by multiplying the output value (Y_{t-1}) of the previous step and the input value (X_t) of this step by the weight of this layer (W_o), and then performing the *sigmoid* operation. Finally, the output value of the LSTM unit is element-wise multiplied by the control information (C_t) after the *tanh* operation and the output of the *sigmoid* layer.

4.3. GRU neural network

GRU (Gated Recurrent Unit) is simplified from LSTM [28,29]. GRU is a variant of LSTM, which is with a simpler structure than LSTM, and has more advantages when building large models. But in this problem, whether GRU has a better effect than LSTM needs to be verified through experiments. Its structure is shown in Fig. 5.

z_t stands for update gate, and its function is similar to the forget gate and input gate of LSTM. It determines what information is discarded and what new information is added. r_t stands for the reset gate, and the reset gate is used to determine the degree to which the previous information is discarded. $\tilde{Y}_t$ is the output gate, which is similar to the role of LSTM. The specific calculation process is shown in the Eq. (7).

$$\begin{aligned} z_t &= \sigma(W_{zy} \cdot Y_{t-1} + W_{zx} \cdot X_t) \\ r_t &= \sigma(W_{ry} \cdot Y_{t-1} + W_{rx} \cdot X_t) \\ \tilde{Y}_t &= \tanh(W \cdot [r_t \cdot Y_{t-1}, X_t]) \\ Y_t &= (1 - z_t) \cdot Y_{t-1} + z_t \cdot \tilde{Y}_t \end{aligned} \quad (7)$$

The calculation method of Eq. (7) is similar to that of LSTM. Y_t is calculated differently. When calculating Y_t , a “1-” operation is performed on the calculated z_t , and the value of z_t is subtracted from 1 and then multiplied by Y_{t-1} .

5. Algorithm

The main idea of the algorithm is to match multiple adapted general models with the six data points measured by the new sensor, and establish a new sensor model through the adapted general model. In this way, a new model can be established after measuring only six data points for a newly produced sensor. Compared with normal calibration, the required measurement time and measurement gas are greatly reduced, so the calibration efficiency is improved and the cost is reduced. And because of the characteristics of the neural network, the new model established has high accuracy at the same time. The types of general models mainly include BP neural network, LSTM neural network and GRU neural network. The difference among the three models is mainly in the types of neurons used in the hidden layer. A type of neural networks model with the best comprehensive performance is selected as the general model using model through experimental comparison in the experimental part.

The main steps of the fast calibration algorithm are as follows:

- Step 1: N uncalibrated sensors are taken to measure group m data points, namely AD value at a certain temperature T and a certain concentration C . (T, AD, C) is taken as a group of data points. The N is larger, the number of model samples will be more. The m is larger, the precision of a single model is higher. Obviously, the algorithm works better when N and m are larger. However, an increase in N and m will greatly enhance the time and financial cost of data collection.
- Step 2: The sequence of m data points collected by each sensor is randomly disrupted, and normalization is carried out. The specific normalization method is shown in the following section.
- Step 3: For each sensor, 20% of the data points are randomly selected as a validation set training model using neural network model training. And N neural network models are obtained.
- Step 4: For the newly produced sensor to be calibrated, AD values at the two concentration points of 400 ppm and 2000 ppm are measured at each temperature point (i.e. 15 °C, 25 °C, 35 °C).
- Step 5: We perform the following operations on N neural network models generated in Step 2: firstly, we normalize the six data points generated in Step 3, and then substitute N neural network models for calculation (the normalization method is shown in the next section). The output value of the model is obtained by calculation. CO2 concentration is acquired by the inverse normalization of the output value. The population standard deviation of the difference between the six predicted values and the true values is gained by calculation.
- Step 6: We select multiple from N model overall model as models of the minimum standard deviation estimation model. The output values of these models are calculated and the average value of these output values is taken as the true output value.

6. Experiment and result analysis

6.1. Data collection

In the early data collection stage, we used air and carbon dioxide to match the carbon dioxide concentration value in the gas, but the measured data had errors, which led to certain deviations in data analysis and model building. Later, we used pure CO₂ gas and pure N₂ gas to carry out gas distribution and data acquisition in a professional temperature box. We were using a high-performance new production of carbon dioxide sensor as the test object. We have collected forty points, temperature from 0 °C to 45 °C, it is a temperature point for every 5 °C, and there is a total of ten temperature points; Data of four concentration points (i.e. 400ppm, 1000ppm, 2000ppm, 4000ppm) were collected at each temperature point.

The specific steps of collection are as follows:

- Step 1: We put the sensor into the corresponding temperature box and set the target temperature in the calibration system. The temperature in the temperature box will be constant in about an hour.
- Step 2: According to the target concentration value, gas distribution is carried out in the temperature box. After 30–40 min, the gas concentration stabilizes, and a measurement operation is completed.
- Step 3: Step 2 is repeated until the AD value of the sensor at all concentration values is recorded at the specified temperature point.
- Step 4: The procedures of Steps 1 through 3 are repeated until the sensor AD values at each concentration at all temperature values are recorded at regular intervals of 5 °C between 0 °C and 45 °C.
- Step 5: We process the obtained experimental data and acquire eight thousand data points from it as the data set for model training and testing. The specific process is as follows.

Table 1
Source data sample.

C (ppm)	T (°C)	AD value
400	6.0	30835
1000	6.0	30215
2000	6.0	29491
4000	6.1	28554
400	11.2	31131
1000	11.2	30489
2000	11.2	29742
4000	11.2	28771

A total of 450,000 raw data from 200 sensors are collected from the test. After analyzing the data, we extract the stable measurement data (the output voltage tends to be flat or fluctuate up and down in a certain interval). Then we extract 5 data from the stable data of each measurement and take the average value which is the data point of the current temperature and current concentration of the sensor. Each sensor is collected forty data points, there is a total of eight thousand data points. Finally, We use 70% of the sensors as the data source (i.e. 140 sensors) needed for model building and use the remaining 30% of the sensors as the source of verification data (i.e. 60 sensors).

The accuracy and effectiveness of the collected data are guaranteed through the collection of large amounts of data, complete experimental equipment and precise gas ratio. The specific data is shown in Table 1 (only part of the data is shown in the table). Because the temperature in the sensor cavity is different from the ambient temperature in the temperature box, the temperature in the sensor cavity will be higher than the ambient temperature, and what actually affects the output of the sensor is the temperature in the sensor cavity, so only the temperature in the sensor cavity is in this table.

6.2. Data normalization

The sensor is designed to have a temperature range of 0 °C to 45 °C and a CO₂ concentration range of 400ppm to 4000ppm.

Firstly, the selected six data points only contain three temperatures (i.e. 15 °C, 25 °C, 35 °C) and two concentrations (i.e. 400ppm, 2000 ppm). Secondly, it is impossible to obtain a large number of data points at different temperatures and concentrations at one time during actual use. Therefore, during calibration, we do not know the maximum and minimum of the actual value of the new sensor. So, during normalization, only six data points can be used to predict the theoretical maximum and minimum values.

For the temperature in the sensor cavity, we normalize it by the following method. We obtain the temperature in the cavity when the ambient temperature is 35, and add 10 to this temperature, and the obtained value is the estimated value of the maximum temperature in the sensor cavity. Similarly, we obtain the temperature in the cavity when the ambient temperature is 15, and minus the temperature by 15, and the obtained value is the estimated value of the minimum temperature in the sensor cavity.

For AD value, we normalize it by the following method. We first obtain the AD value when the ambient temperature is 35 and the concentration is 400 ppm. And then we are going to add 500 to this value, which is the maximum estimated value of AD value. Similarly, we first get the AD value of the data point when the ambient temperature is 15 and the concentration is 2000 ppm. And then we subtract 2500 from this value, which is the minimum estimated value of AD value. The difference between the AD value at 35 °C and 400 ppm and the actual maximum AD value of each sensor is calculated. Then, the statistic of 500 is obtained by selecting the maximum of the difference and increasing the maximum appropriately. Similarly, the difference between the AD value at 15 °C and 2000 ppm and the actual minimum AD value of each sensor is calculated. Then, the statistic of 2500 is obtained by selecting the maximum of the difference and increasing

the maximum appropriately. We draw some conclusions from the feasibility analysis experiment in the early stage. Due to the different characteristics of different sensors, if the theoretical maximum value is used for normalization, it may lead to comparatively large differences in the general model, which is not conducive to the establishment of the general model. When the new sensor is normalizing, the actual maximum and minimum values may exceed the theoretical maximum and minimum values. We limit the actual maximum and minimum values to the theoretical maximum and minimum values, because the exceeded value will not actually be too large.

For label value, we use 4000 ppm as the estimated maximum value and 0 ppm as the estimated minimum value.

6.3. Model building and selecting

We randomly select 140 sensor data sets to ensure the randomness of model samples. The normalized input value and label value are used to train the neural network models. After 2000 iterations, the neural network model of the sensor is obtained, and the model is saved to use later.

The data set of the remaining 60 sensors are used as the test data set. We select six data points from the data set including three temperature points (i.e. 15 °C, 25 °C, 35 °C) and two concentrations (i.e. 400 ppm, 2000 ppm). These data points are reference data points for selecting the model. The main purpose of choosing these six data points is to save costs. Because increasing or decreasing the temperature and increasing the concentration will increase the cost. And in actual use, the focus is on the accuracy of carbon dioxide concentration measurement at room temperature. The above data points are selected after comprehensive consideration.

For model selection, firstly, the normalized six data points are used as the input value of the model, and the output value of the model is obtained. Secondly, the output values are denormalized to obtain the predicted value of CO₂ concentration. Then, the population standard deviation between the six predicted values and the true value are obtained by calculating. Finally, multiple models are chosen with the smallest population standard deviation and these models are the estimated model of the new sensor. The average of these output values is the true output value. Using the average of these models as the output value can reduce the individual impact of a single model, and make the established model closer to the actual model of the sensor to be calibrated.

This paper uses three neural network models (i.e. BP, LSTM, GRU) to compare the results. The comparison results are shown in Fig. 6, Fig. 7, and Fig. 8. According to Fig. 6, comparing the data point qualified rates of the three models, the average data point qualified rates of the three models are similar, and they are all around 90% in the range of 5–10 hidden layer nodes. According to Fig. 7, comparing the time required to establish a model of the three models, the average time consumed by a single model of the BP neural network is 2.197 s, which is much lower than the other two models. According to Fig. 8, comparing the time consumption of the model selection and the qualified rate verification of the three models, the average time consumed by a single sensor of the BP neural network is 18.402 s, which is less than the other two models.

According to the theory, BP neural network is the simplest neural network model among the three neural networks. Although the structure of the BP neural network is simple, its effect can meet the requirements of this paper. After experimental comparison, not only its data point qualified rate is slightly higher than the other two models, but also its time consumption is lower than the other two models, and it has higher accuracy and efficiency.

Therefore, we finally decide to use BP neural network as the neural network model used in this paper after experimental comparison. According to the experimental results, the number of hidden layer neurons has little effect on the time consumed for model establishment, model matching, and qualified rate verification in the range of 5–10 hidden layer nodes. Finally, the number of hidden layer nodes is selected as 7 according to the data point qualified rate.

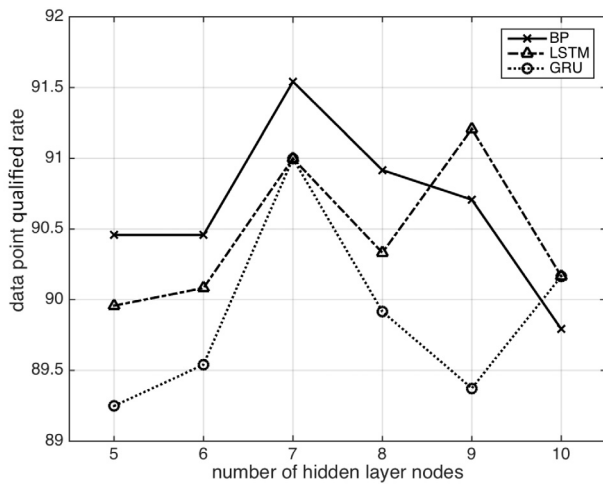


Fig. 6. BP, LSTM, GRU data point qualified rate comparison.

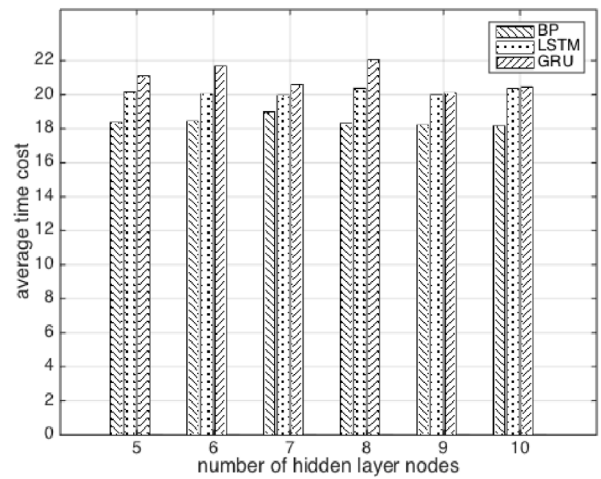


Fig. 8. BP, LSTM, GRU model selection and verification of qualified rate time consumption comparison.

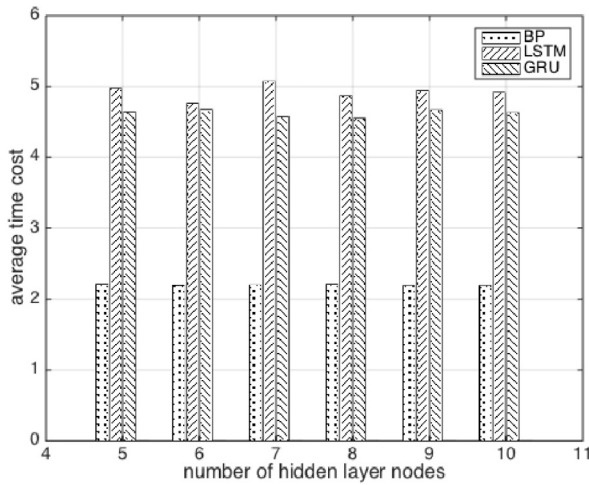


Fig. 7. BP, LSTM, GRU model building time consumption comparison.

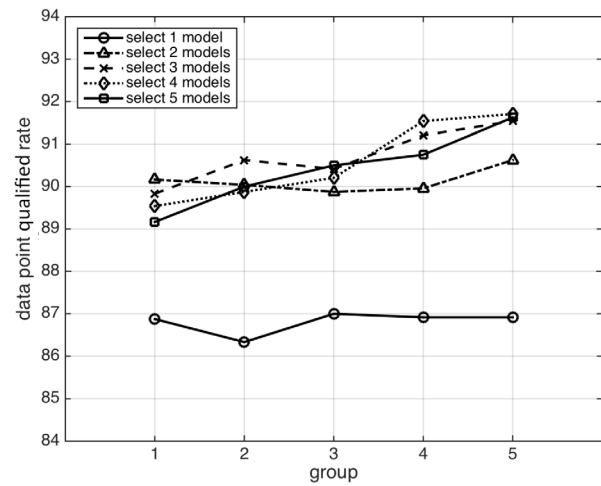


Fig. 9. Model matching experiment results.

6.4. Results

For the same number of test sensors, the more the number of reference models we have, the better the effect of model building we get. Because the more the number of reference models, the more probability of the existence of suitable models. Therefore, the algorithm will have a higher data point qualified rate and a better effect.

The experimental grouping is shown in Table 2 and the result is shown in Fig. 9. Fig. 9 shows that Group 5 has the highest data point qualified rate compared with other groups and the method of selecting multiple models. And selecting multiple models to establish a new model has significantly better effects compared to selecting a single model as the model of the new sensor. It means that this method can eliminate the individual characteristics of a single model to some extent, and make the established model closer to the actual model of the new sensor. And the effect of the algorithm will be better if the number of approximate models selected increases. But there is also an upper limit, and the time cost will be more, so the number of approximate models selected cannot be increased without a limit. The qualified rate of data points basically reaches the peak in the case of using Group 5 and selecting three approximate models at the same time. And choosing more models does not have a relatively large improvement. Therefore, it finally sets the number of the selecting models to three in our paper.

Table 2

Experimental grouping of the number of models and the method of selecting models.

Group	Number of models	Number of test sensors
1	60	60
2	80	60
3	100	60
4	120	60
5	140	60

The current experimental parameters are set to Group 5 while selecting three approximate models. The new models of each sensor are used to calculate forty data points for the corresponding sensor, and the output value is used to calculate the predicted value and deviation rate of each data point. The result is shown in Table 3. It can be seen from the table that the output value predicted by the model has some deviation from the actual concentration. This deviation determines the quality of the new model. The industry believes that the data point can be considered qualified when the data point deviation is floating up and down by industry-standard 5% plus or minus 50 ppm calculation. For example, if the actual concentration of a data point is 400 ppm, then an error of 70 ppm can be allowed according to industry standards. That is, the concentration of the data point can be considered qualified if it is between 330 ppm and 470 ppm.

Table 3
Comparison of actual concentration and predicted concentration.

Ambient temperature (°C)	T (°C)	AD value	True concentration (ppm)	Predicted concentration (ppm)	Deviation rate (%)
0	3.4	44448	400	409.63	2.41
0	3.6	43749	1000	957.88	4.21
0	3.5	42954	2000	1982.16	0.89
0	3.6	41937	4000	3986.48	0.34
5	8.6	44709	400	395.50	1.12
5	8.6	44011	1000	972.97	2.70
5	8.6	43210	2000	2004.72	0.24
5	8.6	42177	4000	3983.20	0.42
...	...	...	...	...	...

Table 4
Qualified rate of data points at each temperature.

Ambient temperature (°C)	Data point qualified rate (%)
0	64.166
5	80.417
10	95.000
15	96.667
20	98.333
25	98.750
30	100.000
35	99.167
40	96.250
45	86.667

Table 5
Compare with the data point qualified rate of two competing products.

Ambient temperature (°C)	Fast calibration (%)	Product 1 (%)	Product 2 (%)
5	80.417	79.861	59.722
20	98.333	82.639	86.805
35	99.167	47.222	94.444
Average	92.639	69.907	80.324

Due to the large individual differences of the sensor's sensitive components at high and low temperatures, the models of the sensor are quite different at high and low temperatures. It also leads to the low qualified rate of data points at high temperatures and low temperatures. As shown in Table 4, the qualified rate of data points is relatively poor at 0 °C, 5 °C, and 45 °C. But in the range of 20 °C–35 °C, the qualified rate of data points can reach 99.063%. And the overall data point qualified rate reaches 91.542%. The above results have reached the industry-leading level.

6.5. Comparison with competing products

This paper compares the single-channel carbon dioxide sensors of other companies with our algorithm. Two competing products are selected for this experiment. 48 sensors of each competing product are selected, and a total of nine data points were measured at three temperature points (i.e. 5 °C, 20 °C and 35 °C) and three concentration points (i.e. 400 ppm, 2000 ppm, 4000 ppm) for the sensor. The result is shown in Table 5.

Based on the above calculation of the qualified rate, the average qualified rate of the data points of competitive Product 1 is 69.907%. And the average qualified rate of the data points of the competitive Product 2 is 80.324%. It can be seen from the data in Table 5 that the effect of Product 1 at higher temperatures is extremely poor. At lower temperatures, the effect of Product 1 is similar to our algorithm. In other cases, it has a big gap with our algorithm. Product 2 is extremely poor at lower temperatures. The effect of Product 2 is similar to our algorithm at higher temperatures. In other cases, it has a big gap with our algorithm.

As shown in Table 5, our algorithm increases the qualified rate of data points relative to Product 1 by 22.732% and increases the

qualified rate of data points relative to Product 2 by 12.315%. And our algorithm is more stable and has better results at various temperatures compared with these two competing products. According to Table 4, our algorithm has an excellent effect between 10 °C and 40 °C, and the data point qualified rate reaches 97.738%. The data point qualified rate even reaches 99.063% when the temperature is between 20 °C and 35 °C.

7. Conclusion

In order to save costs while improving calibration efficiency and accuracy, we proposed a fast calibration algorithm for Non-Dispersive Infrared (NDIR) single-channel carbon dioxide sensor based on deep learning. Firstly, we establish N network models which consist of N sensors by collecting m data points from different temperatures and concentrations. Secondly, we collect six data points from a new sensor which are measured at three temperatures and two concentrations. Thirdly, we choose multiple approximate models from N network models based on the matching of the data points. At last, we regard these models as the estimation model of the new sensor to calibrate the sensor concentration. This method eliminates the individual differences of a single model to a certain extent and achieves the purpose of rapid calibration. After comparing three kinds of neural networks and conducting relevant experiments, we chose BP neural network as the model, and set the number of selected models to three. The results show that the floating up and down by industry-standard 5% plus or minus 50 ppm calculation, the qualified rate of our method is up to 91.542% between 0 °C to 45 °C, and the qualified rate even reaches 99.063% between 20 °C to 35 °C. Compared with similar products, the qualified rate of our method in the calibration of carbon dioxide increases by 12.315% and 22.732% respectively.

The algorithm proposed in this paper has a high qualified rate at some temperatures, but the overall qualified rate is relatively lower than it. The main reason is that under high temperature and low temperature conditions, the degree of model adaptation is not high caused by the differences in individual sensor characteristics, which further leads to a low qualified rate of data points. Therefore, in future work, it is necessary to focus on optimizing the adaptation and model construction of the sensor model under high temperature and low temperature conditions, so as to improve the qualified rate of data points at low temperature and high temperature, and further improve the overall performance of the algorithm.

CRedit authorship contribution statement

Keji Mao: Conceptualization, Methodology, Formal analysis, Methodology, Writing – original draft, Project administration, Funding acquisition. **Jinyu Xu:** Methodology, Code writing, Validation, Data curation, Writing – original draft. **Runhui Jin:** Validation, Formal analysis, Investigation, Data curation, Writing – original draft. **Yuxiang Wang:** Methodology, Data curation, Writing – original draft. **Kai Fang:** Conceptualization, Methodology, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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